CASE

Jamal, 19 · sudden severe chest and back pain · history of sickle cell disease · low-grade fever · sat 91% · smear shows sickled erythrocytes · **acute vaso-occlusive crisis**

BIO 004 · WEEK 3 · CASE DEEP DIVE

Blood

A college student doubled over in chest pain, a single amino acid wrong, and a red cell that loses its shape.

PATIENT CASE

Jamal is 19, a college sophomore with known **sickle cell disease (HbSS)**. He arrives in the ED with **sudden, severe pain in his chest and lower back** that started this morning during a workout. He has had pain crises before, "but this one feels different."

On exam he is in obvious distress, breathing fast (RR 24), **low-grade fever (38.1°C)**, oxygen saturation **91% on room air**, scleral icterus. Labs: **hemoglobin 7.2 g/dL** (down from his baseline of 9.0), reticulocytes elevated, indirect bilirubin elevated, LDH elevated. Peripheral smear shows **numerous sickle-shaped erythrocytes** and Howell-Jolly bodies. Chest X-ray reveals a new **infiltrate in the right lower lobe**. The diagnosis is **acute chest syndrome**, a feared complication of sickle cell disease.

A red blood cell is a beautifully simple device: a biconcave disc, no nucleus, packed with hemoglobin. When one amino acid in hemoglobin is wrong, the cell loses its shape under stress, blocks microvasculature, hemolyzes, and triggers everything Jamal is experiencing.

> **GUIDING QUESTION**
>
> What is blood made of, what does each formed element do, and how does a single amino acid change in a single protein make Jamal's erythrocytes deadly to him?

PREWORK

Companion to your Week 3 Blood prework page. Terms, video, and spaced recall for plasma, formed elements, erythrocytes, leukocytes, platelets, hemoglobin, blood types, and hemostasis live there.

1 COMPOSITION

What's in a tube of blood after you spin it

Centrifuge a tube of blood and it separates into three layers:

Plasma (top, ~55%)

Straw-colored fluid. 90% water, plus dissolved proteins (albumin, globulins, fibrinogen), electrolytes, nutrients, wastes, hormones. The transport medium for everything blood carries.

Buffy coat (middle, <1%)

Thin layer of white blood cells (leukocytes) and platelets.

Erythrocytes (bottom, ~45%)

Red blood cells. Pack the bottom of the tube. The fraction (called the hematocrit) is normally 38-50% depending on sex and altitude.

Blood is classified as a *connective tissue* because it has cells (formed elements) suspended in an extracellular matrix (plasma). The fibers (fibrinogen) only become visible when the blood clots.

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2 ERYTHROCYTES

Built for one job, optimized down to the last detail

Red blood cells (erythrocytes) carry oxygen from the lungs to the tissues and carbon dioxide back. They are built for that single job:

Biconcave disc shape

Maximizes surface area for gas exchange and lets the cell flex through narrow capillaries.

No nucleus, no organelles

More room for hemoglobin. Cell lives ~120 days (no DNA, no repair).

Packed with hemoglobin

~250 million hemoglobin molecules per cell. Each hemoglobin = 4 globin chains + 4 heme groups, each heme carrying one iron atom that binds one oxygen.

Flexible membrane

Spectrin cytoskeleton lets the disc deform through capillaries narrower than the cell itself.

What's wrong in Jamal's erythrocytes. A single point mutation in the beta-globin gene changes one amino acid (glutamic acid → valine). The resulting *hemoglobin S* behaves normally when oxygenated. When deoxygenated, HbS molecules polymerize and stack into rigid rods. The biconcave disc deforms into a rigid sickle shape. Sickled cells don't flex through capillaries. They block them. The blockage causes pain (bone, chest, abdomen), ischemia, and infarction. Sickled cells also hemolyze, dropping the hemoglobin level and releasing bilirubin (the scleral icterus).

3 LEUKOCYTES

Five types, two big categories, one immune system

White blood cells (leukocytes) defend against pathogens and clean up damaged tissue. They are divided into two groups:

CELL	GRANULES?	JOB	% OF WBC
Neutrophil	granular	phagocytose bacteria; first responders to acute infection	50-70
Lymphocyte	agranular	T cells, B cells, NK cells. Adaptive and innate immunity.	20-40
Monocyte	agranular	differentiate into macrophages in tissue; chronic infection, debris clearance	2-8
Eosinophil	granular	parasites, allergic reactions	1-4
Basophil	granular	release histamine; allergic reactions	< 1

Mnemonic for relative abundance: *Never Let Monkeys Eat Bananas* (Neutrophil, Lymphocyte, Monocyte, Eosinophil, Basophil).

4 PLATELETS & HEMOSTASIS

Cell fragments that stop bleeding

Platelets (thrombocytes) are cell fragments shed from megakaryocytes in the bone marrow. They have no nucleus. When a blood vessel is injured they aggregate at the site, release signaling molecules that recruit more platelets, and form the initial plug.

Hemostasis (stopping bleeding) has three phases: *vascular spasm* (the vessel constricts), *platelet plug formation*, and *coagulation* (the fibrin mesh). Coagulation is a cascade of clotting factors (intrinsic and extrinsic pathways converging on a common pathway) that culminates in soluble fibrinogen being cleaved to insoluble fibrin, which weaves the platelet plug into a stable clot. Calcium and vitamin K are required cofactors.

5 BLOOD TYPES

ABO and Rh in one paragraph

Erythrocytes carry surface antigens. The two clinically important systems are **ABO** and **Rh**. ABO has four phenotypes (A, B, AB, O) determined by the presence or absence of A and B antigens. Type O has neither (universal donor). Type AB has both (universal recipient). People naturally produce antibodies against antigens they lack (a type A person has anti-B antibodies). Transfusing the wrong type causes hemolytic reaction.

Rh refers primarily to the D antigen. People are Rh-positive (have D) or Rh-negative (lack D). Rh-negative mothers carrying an Rh-positive fetus can be sensitized at delivery and produce anti-D antibodies that attack a subsequent Rh-positive pregnancy (hemolytic disease of the newborn). RhoGAM prevents sensitization.

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6 HEMATOPOIESIS

Where blood cells come from

All formed elements arise from **hematopoietic stem cells** in the **red bone marrow**. In adults, red marrow is concentrated in the axial skeleton, pelvis, and proximal long bones. The stem cells differentiate down two main lines: *myeloid* (erythrocytes, platelets, neutrophils, eosinophils, basophils, monocytes) and *lymphoid* (B cells, T cells, NK cells).

Erythropoiesis is regulated by **erythropoietin (EPO)**, a hormone produced by the kidneys in response to low oxygen. Chronic anemia (like Jamal's) drives the marrow into overdrive — his reticulocyte count is elevated because his marrow is producing erythrocytes as fast as it can to replace the ones that keep being destroyed.

✎ **DRAW ZONE 1** *-- Sketch a spun tube of blood with the three layers labeled. Then sketch a normal biconcave RBC next to a sickled RBC.*

The trick: sickled cells are rigid and don't fold through capillaries. Show one occluding a capillary lumen.

✎ **DRAW ZONE 2** -- *Diagram a small bone marrow showing hematopoietic stem cell differentiating into myeloid and lymphoid lines, ending with the five WBC types, RBCs, and platelets.*

Use a tree branching structure. Star EPO's effect on erythroid differentiation.

RETURNING TO JAMAL

One amino acid, every symptom

Acute chest syndrome in sickle cell disease is the disease at its most dangerous — sickling in the pulmonary microvasculature blocks gas exchange and creates a positive-feedback loop of more hypoxia, more sickling, more occlusion.

"*Sudden chest and back pain*"

Sickled cells occlude microvasculature in bone marrow and chest wall. The pain is from **ischemia and bone infarction**, not muscular.

"*Hemoglobin 7.2 g/dL (down from 9)*"

Acute hemolysis. Sickled cells are fragile; many burst as they squeeze through narrow capillaries. The marrow can't keep up acutely.

"*Elevated reticulocytes, indirect bilirubin, LDH*"

Three signs of hemolysis. Reticulocytes (immature RBCs) flood the blood because the marrow is making replacements as fast as possible. Indirect bilirubin and LDH spike as hemoglobin is broken down.

"*Sat 91%, new RLL infiltrate*"

Acute chest syndrome: sickling in pulmonary capillaries blocks blood flow, causes infarction, and triggers an inflammatory infiltrate. Often started by infection or fat emboli from infarcted marrow.

"*Howell-Jolly bodies on smear*"

Nuclear remnants in RBCs that the spleen normally removes. In sickle cell disease repeated splenic infarcts destroy splenic function; the spleen can no longer clean up. Functional asplenia.

"*Why he's at risk for serious infection*"

Functional asplenia leaves him vulnerable to encapsulated bacteria (S. pneumoniae, H. influenzae, N. meningitidis). All sickle cell patients should be vaccinated against these and may need prophylactic antibiotics.

Erythrocyte, hemoglobin, capillary, hemolysis. Jamal's ED visit is a master class in why the shape of a cell matters. When you go back to the prework, the RBC biconcave disc diagram will land differently — it's the architecture his body fails to maintain.